

Booking Service Database Design

1. Overview

Purpose

The Booking Service is responsible for:

- Managing business availability
- Retrieving available booking slots
- Creating bookings
- Cancelling bookings
- Preventing double-bookings
- Publishing booking events

The Booking Service owns all scheduling and appointment data.

Responsibilities

Availability Management

Configure working hours
Store business schedules
Generate available slots
Retrieve availability

Booking Management

Create bookings
Cancel bookings
View bookings
Validate booking requests
Prevent double-bookings

Event Publishing

BOOKING_CREATED
BOOKING_CANCELLED

2. Database Technology

Property	Value
Database	PostgreSQL
Service	Booking Service
Tables	2
Aggregate Roots	Availability, Booking

3. Why Two Tables?

KevaBook separates:

Availability

Defines when a business can accept appointments.

Example:

Monday 09:00 - 17:00

Tuesday 09:00 - 17:00

Booking

Represents a confirmed appointment.

Example:

June 20

10:00 - 11:00

Booked by:

John Doe

Without this separation:

- Availability becomes difficult to manage.

- Slot generation becomes complicated.
- Booking validation becomes expensive.

Aggregate 1: Availability

Table Name

availability

Purpose

Stores recurring business working schedules.

Table Structure

Column	Type	Required	Description
availabilityId	Long	Yes	Unique primary key
businessId	Long	Yes	Business unique reference key
dayOfWeek	Enum	Yes	Working days
startTime	Time	Yes	Opening time
endTime	Time	Yes	Closing time
isActive	Time	Yes	Availability status
createdAt	Date Time	Yes	Creation date timestamp
updatedAt	Date Time	Yes	Last date updated timestamp

Aggregate 2: Booking

Table Name

bookings

Purpose

Stores confirmed appointments.

Table Structure

Column	Type	Required	Description
bookingId	Long	Yes	Unique primary key
businessId	Long	Yes	Business reference
serviceId	Long	Yes	Service reference
clientName	String	Yes	Customer name
clientEmail	String	Yes	Customer email
clientPhoneNumber	String	Yes	Customer phone number
bookingDate	Date Time	Yes	Booking date
startTime	Date Time	Yes	Appointment start time
endTime	Date Time	Yes	Appointment end time
status	Enum	Yes	Booking status
createdAt	Date Time	Yes	Creation date timestamp
updatedAt	Date Time	Yes	Last update timestamp

Double-Booking Prevention

A business must never have two bookings occupying the same slot.

Example

Allowed:
Business A

10:00

Rejected:
Business A

10:00

Already booked

The database itself becomes the final protection layer.

Data Ownership Rules

Booking Service Owns

Availability
Booking

Booking Service References

Booking service references businessId and serviceId from Business Service and Service Catalog Service respectively. No foreign keys.